MILITARY SPECIFICATION SHEET

ELECTRON TUBE, GAS SWITCHING

TYPE 5927

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: TR, bandpass, frequency range 3,100 to 3,500 MHz, incident power 825 kw

ABSOLUTE RATINGS:

Parameter:	Incident power	Ii	Ebb	Du	Alt
Unit:	kw	μAdc	Vdc	---	ft
Maximum:	---	200	-700	0.001	10,000
Minimum:	100	---	-500	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1
Mounting:	153-JAN, 268-JAN
Mounting position:	Any

TEST CONDITIONS:

Parameter:	Incident power	Ii	tp1	tp2	F	Du
Unit:	kw	μAdc	μs	μs	---	---
Test condition 1:						
Maximum:	---	---	1.15	---	---	---
:	50	---	1.00	---	F2	0.001
Minimum:	---	---	0.85	---	---	---
Test condition 2:						
Maximum:	220	---	1.15	0.60	---	---
:	200	200	1.00	0.50	F2	0.001
Minimum:	180	---	0.85	0.40	---	---
Test condition 3:						
Maximum:	825	---	---	---	---	---
:	750	200	---	---	F3	0.001
Minimum:	675	---	---	---	---	---

Frequency		
F	MHz	±
F1	3,100	0.1%
F2	3,300	0.1%
F3	3,425	0.1%
F4	3,500	0.1%

GENERAL:

Qualifications - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4021	Degradation due to vibration	---	---		---	---	---	---
4473	Low-level VSWR	2, 3	---	$\sigma' = 1.05$	σ	---	1.9	---
4474	High-level VSWR	4, 5	1		σ	---	1.15	---
1011	Humidity	---	---		---	---	---	---
	Quality conformance inspection, part 1	1	---					
4401	Ignitor ignition time	6	---		t	---	5.0	sec
4406	Ignitor voltage drop	6	---	Ii = 200 μAdc	Eid	-250	-400	Vdc
4452	Spike leakage energy	4	2		Ws	---	0.3	erg
4452	Flat leakage power	3	2		pf	---	50	mw
4473	Low-level VSWR	3	---	$\sigma' = 1.05$ (max) F1 and F4 F2	 σ σ	 --- ---	 1.60 1.40	 --- ---
	Quality conformance inspection, part 2							
4401	Dielectric material strain	---	---		---	---	---	---
4416	Insertion loss (fixed tuned)	3	---	F = F2; Ii = 0	Li	---	0.7	dB
4421	Ignitor interaction (insertion loss)	6	---	F = F2; Ii = 200 μAdc	ΔLi	---	0.3	dB
4471	Recovery time	4, 8	3		t	---	15	μs
1036	Bump	7	---	Hammer angle, 25 degrees	---	---	---	---
	Quality conformance inspection, part 3							
---	Life test	4	3	Group D; Ii = 150 μAdc	t	500	---	hrs
---	Life-test end points:	---	---					
4471	Recovery time	4, 8	3		t	---	30	μs
4452	Flat leakage power	3	2		pf	---	50	mW
4452	Spike leakage energy	4	2		Ws	---	0.3	erg
4416	Insertion loss (fixed tuned)	3	---	F = F2; Ii = 0	Li	---	1.2	dB

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NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent, inspection level II.
2. The input standing-wave ratio shall be measured at intervals of 30 MHz from F1 -30 MHz to F4 +30 MHz. At no frequency in this interval shall the voltage standing wave ratio be greater than the amount specified.
3. This test shall be performed using the flanges specified in Drawing 268-JAN.
4. This test shall be performed using the mount specified in Drawing 153-JAN.
5. With a load standing wave ratio of less than 1.05 the voltage standing ratio produced by the tube shall be less than the amount specified.
6. The ignitor power supply shall have an open circuit voltage of 600 Vdc negative with respect to the tube body. The ignitor series resistance shall be 1.60 megohms. The ignitor shall fire within the specified time after application of voltage.
7. Anvil per Drawing 188-JAN.
8. The recovery time shall be measured after 30 minutes of operation under the conditions specified.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Army - MU
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, OS, MC, CG, SH

Preparing activity: Navy - EC

Agent: DSA -ES

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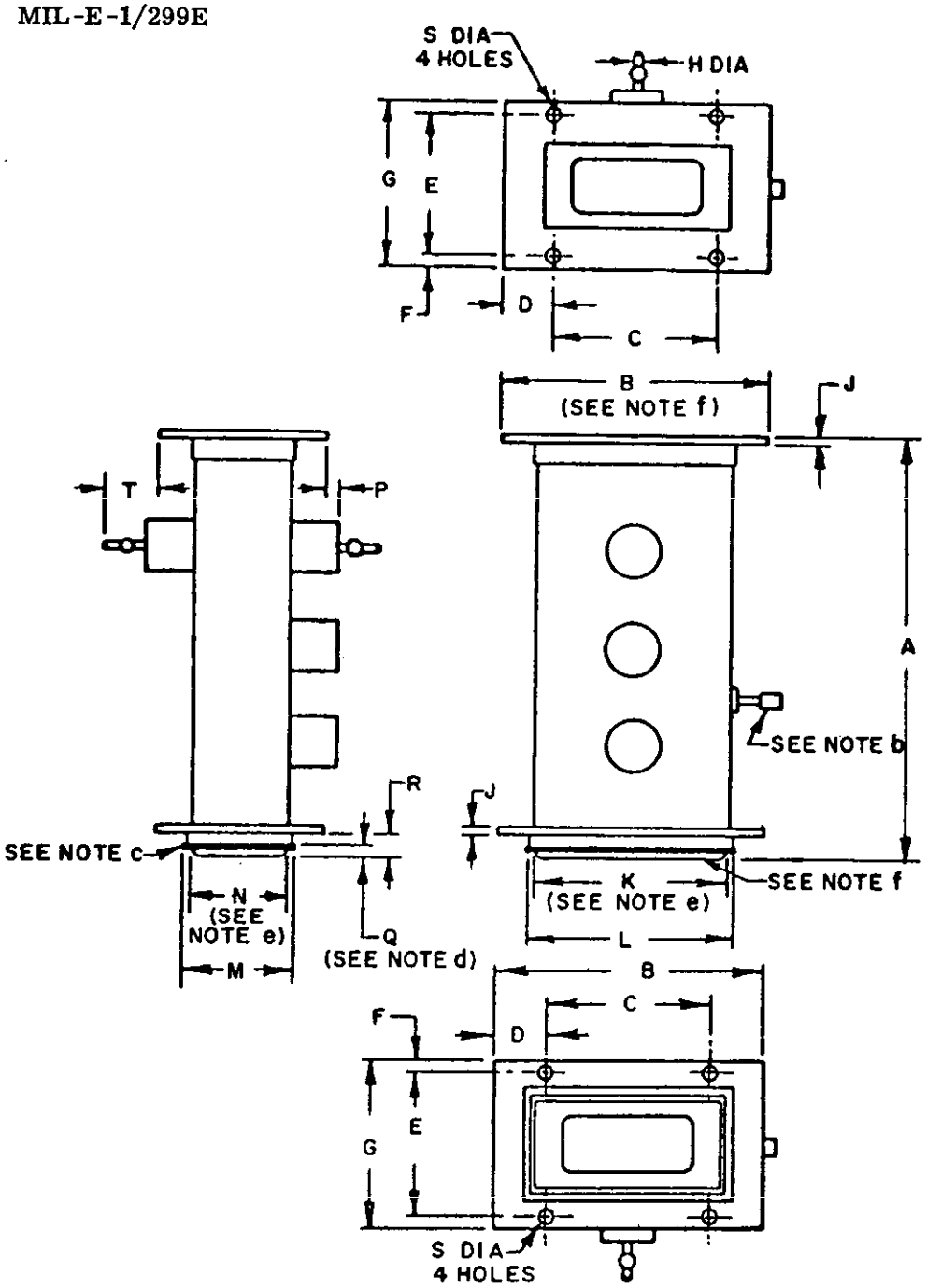


FIGURE 1. Outline drawing of electron tube type 5927.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
	Quality conformance inspection, part 1 (see note a)	
A	4.828 (122.63)	4.888 (124.16)
K		3.008 (76.40)
P		.187 (4.76)
Q	.165 (4.19)	.195 (4.95)
R	.347 (8.81)	.377 (9.58)
T		.812 (20.64)
	Quality conformance inspection, part 2	
C	2.490 (63.25)	2.510 (63.75)
E	2.240 (56.90)	2.260 (57.40)
L	3.203 (81.36)	3.233 (81.85)
M	1.703 (43.26)	1.733 (44.02)
N		1.508 (38.30)
	Reference dimensions	
B	4.125 (104.78)	
D	.812 (20.64)	
F	.187 (4.76)	
G	2.625 (66.68)	
H	.250 (6.35)	
J	.125 (3.18)	
S	.219 (5.56)	

NOTES:
a. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0 percent with inspection level II.
b. Exhaust tube shall not extend beyond flange more than .25 (6.35 mm) inch.
c. Gasket per Drawing 189-JAN. Gasket to be securely attached.
d. Dimension Q shall be measured prior to the attachment of the gasket to the tube.
e. Edges may be rounded.
f. Rhodium flash over silver plating optional.

FIGURE 1. Outline drawing of electron tube type 5927. - Continued

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